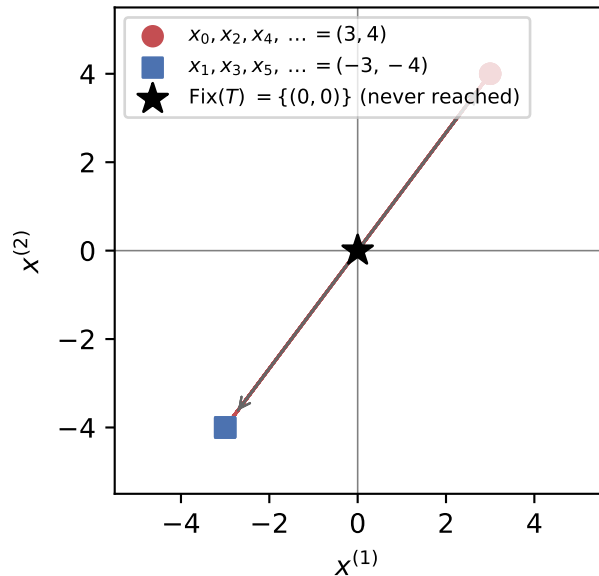


Running example: Picard can fail for nonexpansive maps, but averaging (Krasnoselskii) fixes it

Picard iteration, $T(x) = -x$
(nonexpansive, NOT a contraction):
never converges



Krasnoselskii iteration ($\lambda = 1/2$), $T = P_C$
 $x_{n+1} = \frac{1}{2}x_n + \frac{1}{2}T(x_n)$: converges to $\text{Fix}(T)$

